

# Matthew Butrovich

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## Summary

Database management systems (DBMS) expert with a deep understanding of high-performance data processing—from when queries arrive over the network until their results are durable on disk. Strong background in transactional and analytical DBMS design (query execution, storage management, concurrency control, data formats), Linux internals (eBPF, networking), storage systems (distributed, devices, file systems), performance engineering (benchmarking, observability), and machine learning for systems. Committed to lifelong learning, peer mentorship, and contributing to scientific and open source communities.

## Professional Experience

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|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2024 –      | <b>Apple Inc.</b><br>Senior Distributed Database Engineer <ul style="list-style-type: none"><li>Enhancing Apache Spark’s performance by contributing to Apache DataFusion Comet, a native query execution engine built on Apache Arrow and Apache DataFusion.</li></ul>               | Washington, DC |
| 2009 – 2017 | <b>Western Digital Corporation</b><br>Lead Technician V <ul style="list-style-type: none"><li>Led a team of 17 technicians verifying firmware for HDD and SSD storage devices.</li><li>Promoted four times from initial role as Technician II while achieving B.Sc. degree.</li></ul> | Irvine, CA     |

## Education

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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2019 – 2024 | <b>Carnegie Mellon University</b><br>Doctor of Philosophy (Ph.D.), Computer Science <ul style="list-style-type: none"><li>Advisor: <a href="#">Andy Pavlo</a></li><li>Thesis: “On Embedding Database Management System Logic in Operating Systems via Restricted Programming Environments”</li></ul> | Pittsburgh, PA |
| 2017 – 2018 | <b>Carnegie Mellon University</b><br>Master of Science (M.Sc.), Computer Science                                                                                                                                                                                                                     | Pittsburgh, PA |
| 2013 – 2017 | <b>University of California, Irvine</b><br>Bachelor of Science (B.Sc.), <i>cum laude</i> , Computer Science and Engineering                                                                                                                                                                          | Irvine, CA     |
| 2010 – 2013 | <b>Irvine Valley College</b><br>Certificate of Achievement, Computer Science: Computer Languages                                                                                                                                                                                                     | Irvine, CA     |

## Selected Publications

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>Matthew Butrovich</b> , Samuel Arch, Wan Shen Lim, William Zhang, Jignesh M. Patel, and Andrew Pavlo. “BPF-DB: A Kernel-Embedded Transactional Database Management System For eBPF Applications”. <i>SIGMOD Conference</i> .                                         |
| 2023 | <b>Matthew Butrovich</b> , Karthik Ramanathan, John Rollinson, Wan Shen Lim, William Zhang, Justine Sherry, and Andrew Pavlo. “Tigger: A Database Proxy That Bounces With User-Bypass”. <i>Proc. VLDB Endow.</i>                                                        |
| 2022 | <b>Matthew Butrovich</b> , Wan Shen Lim, Lin Ma, John Rollinson, William Zhang, Yu Xia, and Andrew Pavlo. “Tastes Great! Less Filling! High Performance and Accurate Training Data Collection for Self-Driving Database Management Systems”. <i>SIGMOD Conference</i> . |

## Skills

**Programming Languages:** C, C++, eBPF, Java, Python, Rust, Scala

**Development Tools:** CMake, GDB/LLDB, Git, Google Test, Intel VTune, Make, perf, pytest

## Projects

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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 2024        | <b>BPF-DB</b><br>ACID-compliant embedded DBMS for kernel-space eBPF applications.<br>Role: Sole developer. Wrote eBPF programs in C for storage management (variable length key-value store), transaction management (serializable isolation level), and write-ahead logging. Implemented two sample applications built on BPF-DB: (1) Redis-protocol transactional key-value store that matches state-of-the-art approaches (Dragonfly), and (2) Voter workload comparing serializable transaction performance against VoltDB, achieving 43% higher throughput. |                                                                                     |
| 2023        | <b>Tigger</b><br>PostgreSQL-compatible DBMS proxy based on PgBouncer that bypasses user-space via eBPF.<br>Role: Sole developer. Wrote eBPF programs in C that attach at the socket and Traffic Control layers of the Linux networking stack to apply PostgreSQL protocol logic. This design reduced CPU utilization by up to 42%, and lowered transaction latencies by up to 29%.                                                                                                                                                                               |  |
| 2018 – 2022 | <b>NoisePage</b><br>PostgreSQL-compatible in-memory DBMS designed for autonomous operation.<br>Role: Founding developer responsible for HyPer-style MVCC over Apache Arrow-compatible memory format, PostgreSQL protocol (Simple and Extended Query), and training data collection. Wrote a C++ version of TPC-C to exercise the storage layer for performance and correctness, ported the Bw-Tree from <a href="#">Peloton</a> , and contributed to system catalog design.                                                                                      |  |

## Service

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### To the Open Source Community

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|--------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 2025 – | Committer, Apache DataFusion, <i>The Apache Software Foundation</i> . |  |
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### To the Profession

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|--------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 2025 – | Member, <i>ACM SIGMOD 2026 Program Committee</i> .                         |  |
| 2024   | Member, <i>eBPF Summit 2024 Program Committee</i> .                        |  |
| 2023   | Member, <i>eBPF Summit 2023 Program Committee</i> .                        |  |
| 2023   | Member, <i>ACM SIGMOD 2023 Artifacts &amp; Reproducibility Committee</i> . |  |

### To the University

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|-------------|-------------------------------------------------------------------------------------------------------------|
| 2019 – 2024 | M.Sc. Admissions Committee, Computer Science Department, <i>Carnegie Mellon University</i> .                |
| 2020 – 2024 | Doctoral Review Committee, Computer Science Department, <i>Carnegie Mellon University</i> .                 |
| 2022 – 2024 | Diversity, Equity and Inclusion Committee, Computer Science Department, <i>Carnegie Mellon University</i> . |

## Teaching Experience

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|------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 | <b>15-445/645: Database Systems</b><br>Lecture on join algorithms                       |   |
| 2022 | <b>15-799: Special Topics in Database Systems</b><br>Head teaching assistant            |                                                                                        |
| 2021 | <b>15-445/645: Database Systems</b><br>Teaching assistant                               |                                                                                        |
| 2020 | <b>15-721: Advanced Database Systems</b><br>Head teaching assistant                     |                                                                                        |
| 2019 | <b>15-445/645: Database Systems</b><br>Lecture on two-phase locking concurrency control |   |